ECE362 Junior Design Team 11: Empty Net

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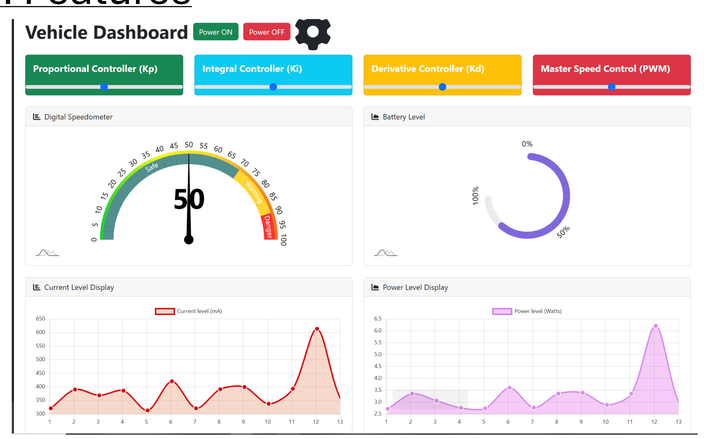
Design For Stakeholders

Team Empty Net is proud to present our design for this competition, an autonomous line following car that will be extremely easy for both the high school teachers and students to pilot and understand

Our design features a website to directly interface with, easily opened with a simple click.
Our hardware includes five supercapacitors that have high design margins so we can guarantee the minimum runtime at max speed is beyond the required competition time.

UI Features



Design features for education

- Website for easy control access
- Github to see code easily
- Dynamic PID control

Design features for safety

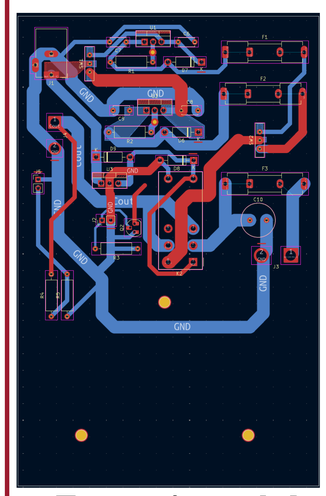
- Current surge protection
- Charge Balancing
- Protection in charging
- Heat protection

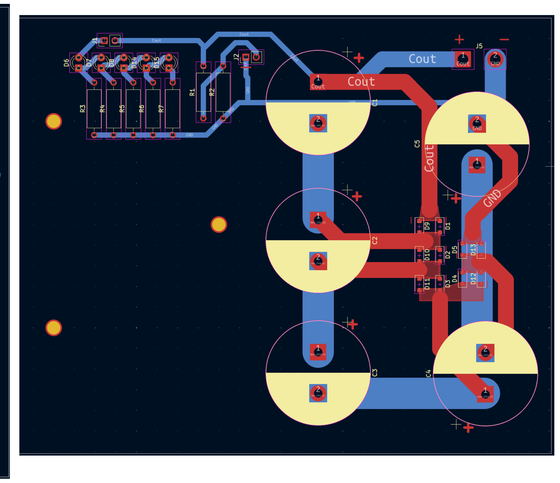
Affordability of design

- **$82.64 out of $100 total budget**
  - Our car can run to higher specs because of this investment in safety and performance

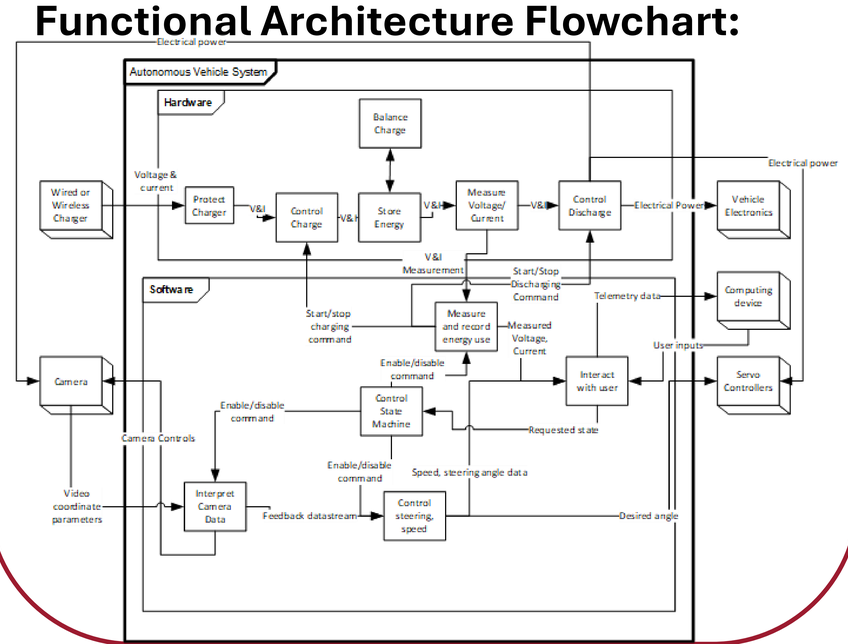
Hardware Design Choices

- Current Limiting circuit implemented using LM317 and LM338 LDO
- Charge Balancing circuit implemented using Zener diodes
- Two separate PCB for capacitor bank and charging control to maximize modularity
- These boards are mounted above the housing for the original battery with a 3D printed bracket and screws for stability





Functional Architecture Flowchart:



Software Design Choices

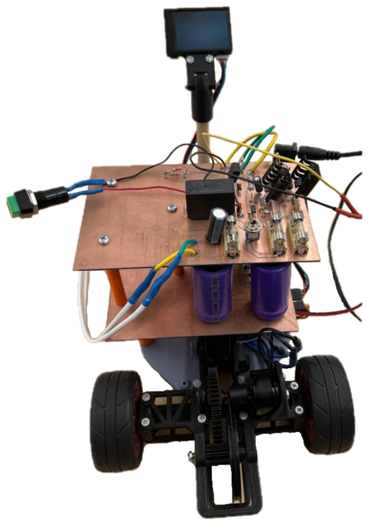
- Line distance from center of camera is where we see the error/when to turn

PID controller

- Used to ensure steering is accurate and efficient from the camera data
- Dynamic change is available

UI software architecture

- We implemented a Firebase website to serve as the controlling and information center of our car
- All code is visible on our Github
- Features:
  - On/Off button
  - Power/Current draw
  - Speed
  - PID variables for steering
  - Battery Level
  - Total energy consumed

Allocation of ESP32 Resources

- This code base uses up approximately 96% of the ESP32 on board memory